ALEX JOSE BRINSON

Physics PhD Student
Massachusetts Institute of Technology

Email: brinson@mit.edu
Web: alexjbrinson.github.io
Phone: (512) 694-9341

Education & Training

Massachusetts Institute of Technology	Cambridge, MA	Physics	Ph.D. 2019-Present
California Institute of Technology	Pasadena,CA	Physics	B.S. 2018

Research & Professional History

2019 – present	Graduate Research Assistant at MIT
2018	Cosmic DAWN Summer Research Fellow
2017 – 2018	Undergraduate Research Assistant, nEDM@SNS
2016	Undergraduate Research Assistant, Keck Array

Publications

1. CREMA: Collinear Resonance for Exotic Molecules and Atoms (preprint). 2026.
2. Nuclear charge radii of aluminium isotopes at the proton drip line (preprint). 2026.
3. Hyperfine structures of $4s, 5s$, $^2S_{1/2}$, $3p\ ^2P_{1/2,3/2}$, and mixed $nd\,y^2D_{3/2,5/2}$ states in ^{27}Al I. *Physical Review A*, 113, 2026.
4. Ionization potential of radium monofluoride. *Physical Review Research*, 8, 2026.
5. Resonance Ionization Spectroscopy Experiment at Facility for Rare Isotope Beams. *Physical Review Research*, 8, 2026.
6. Precision spectroscopy and laser-cooling scheme of a radium-containing molecule. *Nature Physics*, 20, 2024.
7. Observation of the distribution of nuclear magnetization in a molecule. *Science*, 390, 2025.
8. Nuclear Charge Radii of Silicon Isotopes. *Physical Review Letters*, 132, 2024.
9. Isotope Shifts of Radium Monofluoride Molecules. *Physical Review Letters*, 127, 2021.
10. Spectroscopy of short-lived radioactive molecules. *Nature*, 581, 2020.

Technical Skills/Experience

1. **Software:**
   - **Python** - Analysis of spectroscopy data across multiple beamlines and species. Created data acquisition programs for performing spectroscopy on collinear beamline and ion trap. Extensive experience setting up hardware control through custom GUIs and over network communication.
   - **Mathematica** - Numerical and analytical simulations for hardware development, phenomenology of atomic and molecular structure; Data analysis and visualization
   - **COMSOL Multiphysics** - Simulation and analysis of radiofrequency quadrupole (RFQ) trap, using Electrostatics, Charged Particle Tracing, and Molecular Flow modules
   - **CAD** - Designed my thesis experiment, creating parts and assemblies with SolidWorks. Some experience with Inventor.
   - Experience with Javascript, C++, OCaml.
2. **Hardware:**
   - **General** - Participated heavily in construction of the full beamline at EMA Lab at MIT,

as well as the RISE (Resonant Ionization Spectroscopy Experiment) instrument at FRIB. Designed and constructed SCARIEST SCORPIONS trapped ion isotope shift experiment.

- **Optics** - Construction of injection-seeded cavity, frequency doubling and tripling system, and laser ablation ion source. Developed colinear laser spectroscopy paths at 3 facilities (MIT, ISOLDE, FRIB). Set up frequency monitoring and stabilization for various lasers. Designed and assembled fluorescence collection stack. Created a custom laser engraver, using a webcam to correct for open-loop picomotor error, as part of development for an improved laser ablation system.
- **Vacuum Systems** - Designed, simulated, installed, and maintain full vacuum assembly of thesis experiment.
- **Electronics** - Experience building and tuning ion optics. Constructed an affordable thermometry system for our charge exchange cell, using basic voltage divider circuits and an Arduino.

Science Communication and Public Outreach

1. Composed multiple songs describing my research, past and present.
2. Created a music video to publicize results from our Radium Fluoride experiment, which has been viewed over 7000 times by the general public.
3. Given multiple math and physics guest lectures to K-12 classes to educate and inspire the youth.
4. Public Talks:
   - DNP 2018 (Conference Experience for Undergraduates): *Minimizing Environmental Magnetic Field Sources for nEDM*
   - Cosmic DAWN Center 2020 Summer Lunch Talk Series: *Radioactive Molecules: A Low Energy Lab for High Energy Physics*
   - LNS Student Seminar 2022: *Radiofrequency Traps - Operating Principles, Simulations, and Applications in Fundamental Physics"*
   - *DNP 2022: "Development of a Hybrid Multipolar RF trap for Precision Measurements*
   - LNS Student Seminar 2024: *Automated Location-Determinative Ablation, or "How I built a laser engraver"*
   - DNP 2024: *Laser Spectroscopy of Proton-Halo Candidates in Neutron-Deficient Aluminum using the RISE Instrument at BECOLA/FRIB*
   - Quantum Science and Artificial Intelligence for Fundamental Physics 2025: *Isotope Shift Spectroscopy for Nuclear Structure and BSM Physics*
   - LNS Student Seminar 2026: *Isotope Shift Spectroscopy*

Awards

2016 Edward C. and Alice Stone SURF Fellow
2019 MIT Dean of Science Fellow
2021-2022 Helena/Rossi Fellow